Compose files[20]—with some caveats—which can let you sidestep the native task definition format.

Despite this compatibility with Compose, there's no getting around the need to understand ECS's conceptual model, which includes the heady delights of ECS clusters, services, tasks, Application Load Balancers (ALBs) and Elastic Load Balancers (ELBs), VPCs, and more.

Google Kubernetes Engine

Google Kubernetes Engine (GKE)[21] offers a true CaaS model that lets you forget about the underlying hardware. It provides fully managed clusters, meaning that they look after the running of the infrastructure and ensure clusters remain healthy.

Creating a basic cluster is done with a short, clear command—for example:

```
$ gcloud container clusters create guestbook --num-nodes=3
```

This says, "Create new cluster called guestbook with three nodes" and, as you'd expect, GKE handles the creation of the infrastructure for you. There are lots of other possible cluster configurations,[22] such as multiregion clusters.

GKE seems to be the most mature, user-friendly offering in the space—not entirely surprising given that Kubernetes was created at Google.

Amazon Elastic Container Service for Kubernetes

A key drawback of Amazon Elastic Container Service for Kubernetes (Amazon EKS)[23] at the time of writing is its limited availability—currently only in US North Virginia (us-east-1), US Oregon (us-west-2), and US Ohio (us-east-2) regions.[24] However, this will no doubt improve over time.

Unlike GKE, the underlying AWS underpinnings leak through. For example, since EKS needs the ability to create Amazon Elastic Compute Cloud (Amazon EC2)[25] instances on your behalf, you have to create an AWS Identify & Access Management (IAM)[26] service role with the correct permissions in order to get started with EKS.[27]

20. https://docs.aws.amazon.com/AmazonECS/latest/developerguide/cmd-ecs-cli-compose.html
21. https://cloud.google.com/kubernetes-engine/
22. https://cloud.google.com/kubernetes-engine/docs/how-to/creating-a-cluster
23. https://aws.amazon.com/eks/
24. https://aws.amazon.com/about-aws/global-infrastructure/regional-product-services/
25. https://aws.amazon.com/ec2/
26. https://aws.amazon.com/iam/
27. https://docs.aws.amazon.com/eks/latest/userguide/getting-started.html